



COMPENDIUM OF INDIAN
STANDARDS ON

CONCRETE PIPES

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Introduction

The Bureau of Indian Standards (BIS) has established a comprehensive suite of technical standards governing the design, manufacture, testing, and application of concrete and composite pipe systems used in civil infrastructure.

This compendium consolidates and summarizes the key provisions of the principal Indian Standards that apply to various categories of concrete pipes, providing a unified reference for engineers, manufacturers, and project authorities involved in water, sewerage, and drainage infrastructure.

The standards address a diverse range of pipe types and materials suited for specific operational conditions:

- IS 15155:2020 defines the technical requirements for bar/wire wrapped steel cylinder pipes with mortar lining and coating, typically used for water transmission and distribution mains requiring robust structural performance under pressure.
- IS 1916:2018 extends coverage to steel cylinder pipes with concrete lining and coating, accommodating larger diameters (up to 5000 mm) and offering enhanced rigidity for both water supply and sewer systems.
- IS 784:2019 provides specifications for prestressed concrete pipes, including both cylinder and non-cylinder types. These are engineered to counteract internal and external stresses through prestressing techniques, making them suitable for high-pressure conveyance systems.
- IS 458:2021, the most widely used standard, lays down the requirements for precast concrete pipes with and without reinforcement, including pressure and non-pressure varieties. It addresses pipes manufactured by spinning and vertical casting methods and includes detailed classification, material specifications, and strength criteria.
- IS 7322:1985 is a companion standard that specifies specials such as bends, tees, and junctions used in conjunction with reinforced concrete cylinder pipes, ensuring integration and continuity within pipe networks.
- IS 7319:1974 and IS 4350:1967 focus on perforated and porous concrete pipes, respectively. These pipes are essential in under-drainage and subsoil drainage applications where controlled water movement is necessary.

Together, these standards form a complete framework supporting the lifecycle of concrete pipe infrastructure—from planning and design through to manufacturing, quality assurance, and field application. They reflect current engineering practices, incorporate safety margins for performance under varying loads and environmental conditions, and promote uniformity in public and private sector projects across the country.

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IS 458: 2021

PRECAST CONCRETE PIPES (WITH AND WITHOUT REINFORCEMENT) – SPECIFICATION

Scope:

This standard covers the requirements for reinforced and unreinforced precast cement concrete pipes, of both pressure and non-pressure varieties used for water mains, sewers, culverts and irrigation. The requirements for collars are also covered by this standard. Prestressed concrete pipes and pipes with non-circular section are not covered by this standard.

Key Provisions:

Classification:

- a) Unreinforced Concrete Pipes – Class NP1, NP2 and NP3 (non-pressure pipes)
- b) Reinforced Concrete Pipes – Class NP 2, NP 3 and NP 4 (non-pressure pipes) and P-1, P-2 and P-3 (pressure pipes).

Manufacturing process: Spinning and vibration cast

Design, manufacture and testing:

It prescribes raw material requirements including cement, aggregates, reinforcement, mortar, water, rubber rings, and admixtures. Design parameters include barrel thickness, reinforcement details, and end configurations. The standard mandates comprehensive testing methods for strength, permeability, and dimensional conformity.

Uses: Water Mains, sewer, irrigation and culverts/cross drains

IS 784: 2019

PRESTRESSED CONCRETE PIPES (INCLUDING SPECIALS) — SPECIFICATION

Scope:

This standard covers the requirements of prestressed concrete pipes (including specials) with nominal internal diameter in the range of 200 mm to 2500 mm, in which permanent internal stresses are deliberately introduced by tensioned steel to the desired degree to counteract the stresses caused in pipe under service.

Key Provisions:

Classification:

- a) Prestressed Concrete Cylinder Pipes
- b) Prestressed Concrete Non-Cylinder Pipes.

Design, manufacture and testing: It specifies raw materials like cement, steel (both prestressed and untensioned), joint rings, and protective coatings. Detailed guidelines are provided for prestressing, curing, and joint construction. The standard mandates tests including design proving, hydrostatic tests, permeability checks, absorption tests, and dimensional inspections.

Uses: For Water supply, sewerage and culverts

IS 15155:2020

BAR / WIRE WRAPPED STEEL CYLINDER PIPES WITH MORTAR LINING AND COATING (INCLUDING SPECIALS) — SPECIFICATION

Scope:

This standard covers the requirements of bar/wire wrapped steel cylinder pipes with mortar lining and coating with nominal internal diameter in the range of 250 mm to 1 900 mm. Pipes of diameter beyond 1900 mm may be supplied by mutual agreement between purchaser and manufacturer.

Key Provisions:

Design, manufacture and Testing:

It outlines the required raw materials such as cement, aggregates, reinforcement, steel cylinders, and rubber rings. The design focuses on steel wrapping, mortar application, and curing. It includes test methods like hydrostatic pressure tests, mortar strength tests, and permeability tests.

Uses: Water transmission mains, distribution feeder mains, sewer pumping mains, water intake and discharge lines and plant piping

IS 1916:2018

STEEL CYLINDER PIPES WITH CONCRETE LINING AND COATING – SPECIFICATION

Scope:

This standard lays down the requirements for steel cylinder pipes with concrete lining and coating having nominal internal diameter from 200 mm to 5000 mm for use in water mains, sewers, irrigation works and similar situations.

Key Provisions:

Classification: Based on Test pressure

- a) Class 1 (0.5 MPa or 50 m head)
- b) Class 2 (1.0 MPa or 100 m head)
- c) Class 3 (1.5 MPa or 150 m head)
- d) Class 4 (2.0 MPa or 200 m head)
- e) Class 5 (2.5 MPa or 250 m head)
- f) Special Class (Above 2.5 MPa or above 250 m head)

Design, manufacture and testing:

Required materials include steel cylinders, reinforcement, concrete, and mortar. Design provisions include cylinder thickness, cage reinforcement, and cover details. Testing includes hydrostatic and compressive strength tests.

Uses: Water Mains, sewer, and irrigation works

IS 4350:1967

SPECIFICATION FOR CONCRETE POROUS PIPES FOR UNDER DRAINAGE

(Reaffirmed in 2022)

Scope:

This standard lays down the requirements for porous pipes made of concrete for use in under drainage. The requirements cover pipes ranging from 80 mm nominal internal diameter to 900 mm nominal internal diameter with three types of joints.

Key Provisions:

Manufacture, finish and testing:

It mandates the use of specific cement and aggregates to maintain pipe porosity while ensuring structural integrity. Manufacturing guidelines include forming non-porous inverts if necessary. Tests cover dimensional accuracy, load performance, and infiltration characteristics.

Uses: Subsurface drainage, under-drainage systems (e.g., irrigation, roads, sports fields)

IS 7319: 1974

SPECIFICATION FOR PERFORATED CONCRETE PIPES

(Reaffirmed in 2020)

Scope:

This standard lays down the requirements for perforated non-reinforced concrete pipes for use in underdrainage work.

Reinforced cement concrete perforated concrete pipes may be supplied by mutual agreement between the purchaser and the supplier.

Key Provisions:

Classification:

- a) Circular Perforations Pipes
- b) Slotted Perforations Pipes

Design, manufacture and testing: It outlines raw material and size specifications, and prescribes design details for perforations and thorough curing. Required tests include compressive strength, dimensional checks, and absorption rates.

Uses: Underdrainage work in infiltration galleries, reclaiming waterlogged areas and for similar other purposes.

IS 7322:1985

SPECIALS FOR STEEL CYLINDER REINFORCED CONCRETE PIPES

(Reaffirmed in 2020)

Scope:

This standard lays down requirements and methods of tests for steel cylinder reinforced concrete specials for steel cylinder reinforced concrete pipes conforming to IS1916 having nominal internal diameter from 200 to 1800 mm. This specification covers specials having:

- a) spigot and socket ends,
- b) plain ends or slip-in type ends suitable for field welding, and
- c) flanged ends for connection with valves and accessories.

Key Provisions:

Classification:

- a) Same as that given for IS 1916

Design, manufacture and testing:

It defines the material requirements for cement, aggregates, steel, and reinforcement. Design guidelines include cage reinforcement, steel plate thickness, and concrete curing. Tests include compressive strength, dimensional accuracy, and dye penetration to ensure quality.

Uses: curves, bends, branches, manholes, air valves, blow-offs and connection to main line valves and other pipes.

STANDARD ON TEST METHODS

IS 3597: 1998

CONCRETE PIPES – METHODS OF TESTS

(Reaffirmed in 2023)

Scope:

This standard covers methods for carrying out the following tests on concrete pipes, both reinforced concrete and prestressed concrete and of pressure and non-pressure types to evaluate the properties stipulated in the relevant Indian Standards:

a) Three-Edge Bearing Test:

This test evaluates the load-bearing capacity of the pipe. The pipe is tested under a controlled vertical load using top and bottom bearing blocks. Crack formation (at 0.25 mm) and ultimate load are recorded to assess strength and structural integrity.

b) Absorption Test:

This test determines the water absorption capacity of concrete specimens. The procedure involves drying, soaking, and weighing to calculate moisture uptake as a percentage of dry mass.

c) Hydrostatic Test:

This test assesses the pipe's ability to withstand internal water pressure. Pipes are filled with water, air is expelled, and subjected to increasing pressure. The formation of water beads or leakage indicates failure. The pressure is held based on wall thickness before final assessment.

d) Permeability Test:

This test measures water permeability through the pipe wall. It is conducted on prestressed concrete pipes, bar/wire wrapped steel cylinder pipes with mortar coating, and precast concrete pipes. Water absorption over time is measured using a glass tube apparatus and reported in cm³. Acceptance is based on final permeability values.

e) Straightness Test:

This test checks the longitudinal straightness of the pipe bore using a special gauge. If deviations prevent proper contact of the gauge, the pipe is deemed non-compliant.

CODE OF PRACTICE

IS 783:1985

CODE OF PRACTICE FOR LAYING OF CONCRETE PIPES (Reaffirmed in 2022)

Scope:

This standard provides recommended practices for the laying of precast concrete pipes, including methods for calculating loads based on installation conditions and guidance for proper handling, jointing, anchoring, and testing to ensure structural integrity.

Key Provisions:

Design Considerations:

- Methods for determining loads from fill material, superimposed loads (static or dynamic), and internal water load.
- Use of coefficients and load factors based on installation conditions: trench, positive projection, negative projection, wide trench, and imperfect trench.
- Graphs and formulas are provided to calculate vertical loads and to select appropriate pipe class based on test load requirements.

Construction Guidelines:

Handling and Transport: Emphasizes secure loading, proper offloading using slings and spreader beams, and careful stacking/stringing of pipes.

Trench Excavation: Prescribes minimum working space and guidance for sloped or stepped trench sides.

Laying and Jointing: Recommends techniques for proper pipe alignment, joint sealing, and managing expansion gaps. Laying should preferably proceed uphill.

Anchoring on Slopes: On gradients steeper than 1:6, transverse anchor blocks should be provided with spacing defined in the standard.

Additional precautions are suggested for very steep slopes (e.g., water stops, spacers, and restraining the last pipe laid).

Testing Requirements:

Hydrostatic Pressure Test: Pipes shall be tested with water at a specified pressure. Detailed guidance is given for absorption, allowable leakage, and pressure duration.

Leak Detection Methods: Visual inspection, bar probing, acoustic devices, dyes, and gas tracers (e.g., nitrous oxide) are among the methods outlined for identifying pipe leaks during testing.